

Chipmate!

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The Cores have landed, and AMD's still recovering from shellshock, as their hammers have finally been, well hammered! AMD 64 was a glimpse of the future, and the Core delivers more of the same. Will the trend continue? Well, to know where you're going, you've got to know where you are! Wafers on a plate... The K8 and P8 inside!

The driver of evolution is need. PCs are full-blown entertainment stations, and the term "rig" is used to define that monster lurking inside your cabinet... and it's not a 400-horse V8! Such powerhouses are the realm of Intel and AMD. Their latest Core 2 and AMD 64 architectures are revolutionary, even evolutionary advancements over their earlier NetBurst and K7 cores respectively.

AMD 64s have been dominating the desktop market as far as performance goes for nigh to three years. Intel's infamous NetBurst suffered from a serious case of performance-throttling heat-stroke, and lagged behind on nearly all fronts. Any

The AMD 64s are architecturally leaps ahead of the Athlon XPs of yesteryear. The AMD 64 was designed for two sockets—754 and 939. Socket 754 was reserved for the budget segment, while 939 was reserved for the performance-demanding user; while this dividing line was rather slim, it did exist.

The major difference was in the design of the memory controller. Socket 754 had a single-channel, 64-bit integrated memory controller, while Socket 939's architecture included a dual-channel 128-bit controller. The difference in performance came not only from the memory controller

The Core architecture has a 14-stage pipeline, down from the 31 pipelines that NetBurst processors from the Prescott onwards featured. A classic case of when less is more! The K8 architecture has a 12-stage pipeline, up from its former 10



market needs competition to survive, and it's here, as the Core 2 Duo emerges with significant performance leads over everything in the desktop processor realm! AMD's response is highly anticipated... but that's another story for another time.

The Fearsome Twosome

The processor industry as a whole is moving towards "paral-ism". Dual cores have been available for a while now, and quad cores are expected to make an entry shortly. Both the

type used, but also from the fact that the faster CPUs were all made with 939 pins. The IMC supported up to dual-channel DDR 400; later revisions saw native DDR 500 support.

While Intel's switch from DDR to DDR2 was relatively simple—no processor architectural-level changes were required—for AMD, this got a lot more complex. In case you haven't guessed why, it's due to the fact that the memory controller needed redesigning to support DDR 2.

Architecturally, other than a few tweaks to the memory